Sampling from P

Anytime-Valid UCB (ours)

True Missing Mass, R_t^{unseen}

Unseen failure probability

1.0
0.8
0.6
0.4
0.2
0.0

0
500
1000
1500
2000
2500
3000

Number of sampled sessions, t

Sampling from Q

Anytime-Valid UCB (ours)

True Missing Mass, R_t^{unseen}

Unseen failure probability

1.0
0.8
0.6
0.4
0.2
0.0

0
500
1000
1500
2000
2500
3000

Number of sampled sessions, t

Coverage

Sampling from P

Sampling from Q

nominal $1 - \alpha$

Simultaneous coverage

1.00
0.98
0.96
0.94

0
500
1000
1500
2000
2500
3000

Number of sampled sessions, t

Additional stopping cost of Q

Mean $\tau_Q - \tau_P$

Additional samples, $\tau_Q - \tau_P$

200
150
100
50
0

0.120
0.126
0.132
0.138
0.144

Stopping threshold, ε